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Speech-Based Early Detection of Alzheimer's Disease Using Explainable Machine Learning and Ensemble Methods

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Abstract

Alzheimer's disease is a disease that results in progressive degeneration of the memory and cognitive ability of the patient. Early detection of this disease becomes extremely important in such a scenario. Traditional ways of detecting this disease, such as neuropsychological tests and neuroimaging techniques, are not affordable and require extensive time. Consequently, it would not be possible to carry out such tests on a large scale because of the associated costs and extensive efforts involved. In this research, an alternative strategy based on speech signals will be adopted to develop a faster and cost-effective detection model.

In this case, two reliable ML models, SVM and XGBoost, were applied for pattern recognition among features obtained from the dataset. In order to achieve better results in classification, predictions by both of these algorithms were combined by applying the SX Fusion classifier based on a soft voting scheme. Furthermore, SX Fusion was additionally enhanced through the POS method to improve its performance in terms of classification quality and reliability. To interpret the results of classification based on particular features, the SHAP algorithm was used to estimate their importance and contribution. As seen from the results obtained from the experiment, the enhanced SX Fusion model demonstrates higher efficiency than standalone models in diagnosing Alzheimer's at its early stages based on speech characteristics. Therefore, one can assume that the proposed approach is an effective solution in terms of implementation in the current healthcare infrastructure.

Keywords: Alzheimer's disease; automatic speech recognition; cognitive impairment detection; artificial intelligence; speech acoustics; mel frequency cepstral coefficients (MFCC); support vector machine (SVM); XGBoost; ensemble learning techniques; POS; SHAP; DSP; early diagnosis of dementia; non-invasive screening

Introduction

Alzheimer's disease is an illness that affects cognitive abilities, memory, and language skills. Early diagnosis of the Alzheimer's disease is essential for effective treatment to ensure a high quality of life. Traditional diagnostic methods include neuroimaging, CSF analysis, and cognitive testing. However, according to recent studies, analyzing the patient's speech is a helpful technique to identify whether one suffers from Alzheimer's disease. An individual who suffers from Alzheimer's disease has a peculiar way of speaking. This condition manifests itself through difficulties in speaking, improper speech timing, constant pauses, and word retrieval issues. Speech analysis outperforms traditional diagnostic procedures in numerous aspects. For instance, it does not cause any pain, it is inexpensive, and it does not involve the use of specific medical devices.

The latest advancement in the area of machine learning has enabled the creation of intelligent machines capable of analyzing the speech signal and diagnosing cognitive disorder from the pattern in speech signals. Mel Frequency Cepstral Coefficients (MFCC), Zero Crossing Rate, and Spectral Centroid are some of the features that are useful in characterizing temporal and spectral properties in speech. To solve the above problems, a predictive model for Alzheimer's disease using speech signals has been proposed. The model utilizes the multiple machine learning approaches to improve the accuracy in predictions.

SVM and XGBoost are applied in order to identify significant features from the extracted pool of acoustic features. The output results obtained from both models are combined through soft voting mechanism to get SX fusion model. The performance of SX fusion model can be improved further through POS optimization. In addition to this, SHAP technique is also employed to improve the interpretability of prediction model through feature selection.

In summary, this research paper presents a speech recognition system that consists of acoustic feature extraction, modeling via ensemble algorithms such as SVM and XGBoost, optimizing them through POS methods, and lastly, SHAP to interpret the model in artificial intelligence. Consequently, it can be viewed as a potential solution to detect Alzheimer's disease at its early stages and will serve as an efficient tool in the contemporary digital health sector.

Literature Review

Advancements in artificial intelligence and speech processing techniques have significantly contributed to the creation of automation systems that detect Alzheimer's disease through voice biomarkers.

De Filippis and Al Foysal [1] developed an interpretable method that utilizes linguistic biomarkers extracted from voice samples in order to diagnose Alzheimer's disease. While the importance of an interpretable biomarker model is discussed by the researchers, the focus of their biomarker model was primarily linguistic rather than acoustic.

Kiyoshige et al. [2] proposed an AI-driven voice biomarker system for early cognitive impairment detection. It is stated in the study that speech signals carry considerable value for diagnostics, though the complexity of the model construction is notable.

Rezaii et al. [3] evaluated the usefulness of voice analysis in the early detection of Alzheimer's and concluded that small changes in one's speech can be helpful in the process of diagnosing cognitive impairment.

Sharafeldeen et al. [4] conducted a review on the application of machine learning methods for Alzheimer's diagnosis through speech analysis. The study offers knowledge about features extraction techniques such as MFCC and the issue of generalization and limited datasets during training.

Oiza-Zapata et al. [5] published a paper on the topic of explainable artificial intelligence approaches for speech-based Alzheimer's disease detection. They stressed the importance of interpretable models for health care applications. The study found that feature explanation enhanced user confidence.

There is a lot of research focusing on the utilization of deep learning models and foundation models in Alzheimer's disease detection. For example, Park et al. [6] proposed a reasoning method with large language models, whereas Li et al. [7] examined foundation speech models that performed better than traditional methods. However, they needed vast amounts of training data and high computational costs.

Limitations

There are several limitations that can be identified based on the review of existing literature. The application of deep learning or multi-modal approaches [6, 10] requires massive datasets as well as substantial computational resources. While more recent methods perform better in comparison to older ones, their application is often non-interpretable and difficult to scale [5]. Another issue is that light-weight interpretable approaches utilizing acoustic features have not been investigated in much detail yet.

Proposed Approach

This paper attempts to address these shortcomings by presenting a lightweight and explainable speech-based system for Alzheimer's diagnosis. The proposed system is developed based on the extraction of acoustic features using ensemble learning and explainable AI models.

Research Gap

The following are some of the issues associated with current methodologies involved in the use of speech analysis for the diagnosis of Alzheimer's disease. They employ single model techniques, have limited generalization on multiple datasets, and have inadequate feature representation. The majority of the existing models tend to ignore temporal information or focus more on the model's complexity than representation. One other disadvantage of using current methodologies is the existence of many black-box models that are less interpretable. These methodologies are computationally expensive and thus not suitable for real-time applications.

Proposed Solution

Since all the problems raised above have already been addressed in this research, the proposed method for detecting Alzheimer's disease through speech processing by means of acoustic feature extraction and ensembling technique can be considered. Specifically, some acoustic features such as Mel-Frequency Cepstral Coefficients, Zero Crossing Rate, and Spectral Centroid have been chosen to ensure that both spectral and temporal properties would be considered. Two base models were chosen to build the presented approach – SVM and XGBoost; the first model focuses on establishing decision boundaries that cannot be linear, while the latter one helps capture relationships among features. As for the way to combine the results from different classifiers, the soft voting strategy was applied.

Methodology

The proposed method for early detection of Alzheimer's disease using speech signal processing and machine learning techniques is as follows: The whole methodology can be considered in five distinct stages, namely,

i. Data acquisition and preprocessing,
ii. Feature extraction,
iii. Normalization of features,
iv. Classification using machine learning techniques, and
v. Decision level fusion.

System Architecture

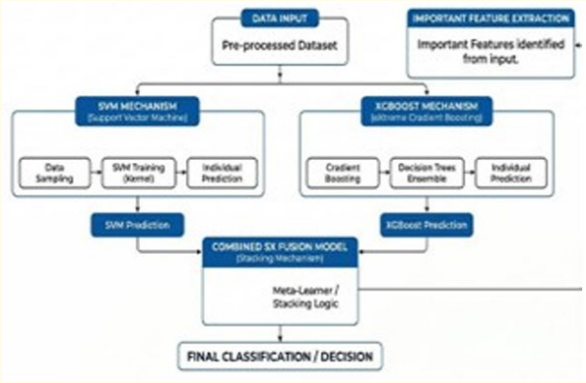


Figure 1: Speech-Based Alzheimer's Detection System Architecture.

Figure 1 demonstrates the complete structure of the proposed system. The system receives speech inputs, analyzes acoustic features, processes with classification algorithms, and produces results.

Data Preprocessing

The speech signals are captured and classified into two classes – dementia and non-dementia classes. Preprocessing includes standardizing the length of the signal within a specific time window, along with keeping the sampling conditions same. Let the input speech signal be denoted by $x(t)$. Then, the input signal $x(t)$ is trimmed to produce a speech signal of a constant length T.

Feature Extraction

The feature vectors are then derived from the acoustic properties associated with the cognitive deterioration in the voice signal. The set of features F can thus be formulated as:

$$F = [MFCC_1, MFCC_2, \ldots, MFCC_{13}, ZCR, SC] \quad (1)$$

Where $MFCC_i$ is the ith Mel Frequency Cepstral Coefficient, ZCR is the zero-crossing rate, while SC represents the spectral centroid. These features are extracted through short-term processing, with the feature extraction being performed by

Feature Normalization

Standardization is used to scale feature vectors uniformly as below:

$$X' = \frac{X - \mu}{\sigma} \quad (2)$$

Here μ represents the mean while σ represents the standard deviation.

Models for Classification

Support Vector Machine (SVM): An SVM model using an RBF kernel is employed to model a nonlinear decision boundary function which is as below:

$$f(x) = \sum_{i=1}^{n} \alpha_i y_i K(x_i, x) + b \quad (3)$$

Here $K(x_i, x)$ represents the kernel function.

Extreme Gradient Boosting (XGBoost): The XGBoost model is applied to analyze complex interactions among the variables using an ensemble of decision trees. The formula for the model output is provided by:

$$\hat{y} = \sum_{k=1}^{n} f_k(x) \quad (4)$$

where f_k represents.

Ensemble Learning (Decision-Level Fusion): To improve robustness, the use of a soft voting method technique is applied. Calculation of the final probability is carried out using the formula below:

$$P_{\text{final}} = \frac{P_{\text{SVM}} + P_{\text{XGB}}}{2} \quad (5)$$

parts of speech

POS-Based Optimization

The POS technique enhances the performance of the ensemble classifier through the continuous optimization of the particles that indicate the potential solutions. This results in the minimization of the classification error rate and increased efficiency through as-

sessing the fitness of the particles based on their accuracies.

Model Evaluation

The efficiency of the system is evaluated according to the traditional metrics, such as accuracy, precision, recall, and F1 score. The confusion matrix is built to evaluate the efficiency of classification for each class separately.

Integration of Explainable AI

XAI methods, such as SHAP (SHapley Additive explanations), are embedded into the proposed system to increase its transparency and explainability. Integrating the described technique into the XGBoost model helps investigate the influence of each acoustic feature on the result.

- ***Feature Importance:*** This aspect decides which features play an important role in classification.
- ***Model Interpretation:*** It gives insight into the effect of changes in feature values on the output of the model.

Results and Analysis

Model Performance Evaluation

Performance analysis of the proposed model is done through confusion matrices and accuracy metrics for the following algorithms: SVM, XGBoost, and ensemble model.

SVM Performance: The effectiveness of the Support Vector Machine (SVM) is determined from the confusion matrix given in Fig. 2 below.

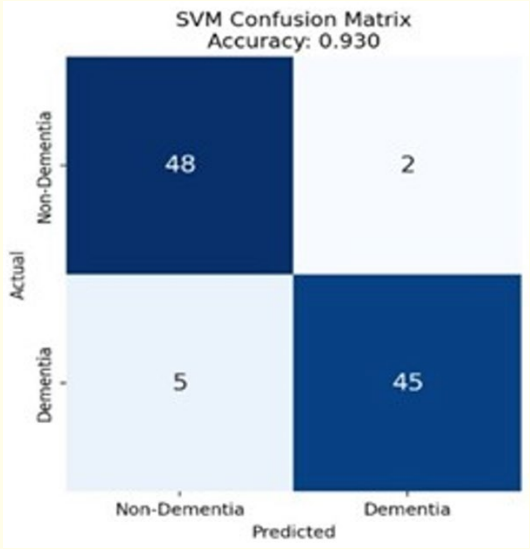


Figure 2: Confusion Matrix of SVM Model.

From the confusion matrix above, it is clear that the classifier has effectively classified 48 non-dementia patients and 45 dementia patients, while incorrectly classifying 2 non-dementia patients as being dementia and 5 dementia patients as being non-dementia. Its accuracy is, therefore, 93%, which is quite high indicating excellent performance of the algorithm. However, the main weakness of the SVM classifier is poor recall since it cannot classify some of the dementia patients.

XGBoost Performance: Performance of the XGBoost model can be analyzed by using the confusion matrix presented in Fig. 3.

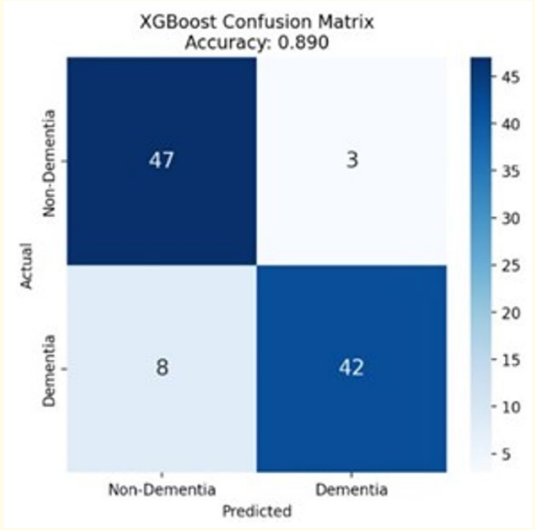


Figure 3: XGBoost Model Confusion Matrix.

From the above table, it can be seen that the XGBoost model can classify 47 non-dementia patients and 42 dementia patients correctly, while misclassifies 3 non-dementia patients and 8 dementia patients. It has improved performance in terms of classification of non-dementia patients when compared with SVM models; nevertheless, due to a large number of errors for dementia patients, the accuracy percentage for XGBoost is lower than that for SVM at 89%. Despite its ability to handle complex relationships among different variables, it is not possible to employ it for the diagnosis of early-stage Alzheimer's disease.

Ensemble(SX Fusion) Model Performance: Performance Evaluation of the Ensembling Method (SX Fusion).

The performance evaluation of the SX fusion model is done by using the confusion matrix shown in Fig. 4.

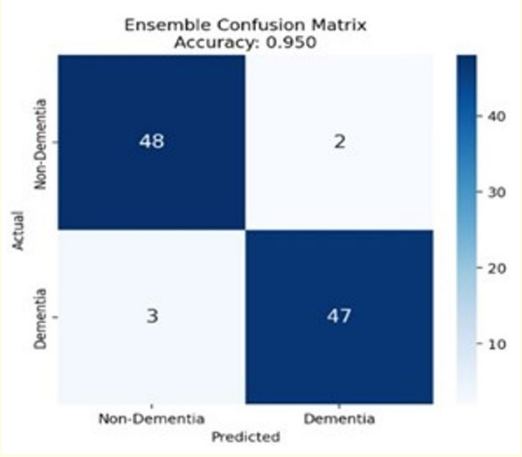


Figure 4: Confusion Matrix of Ensemble Model.

Though the ensemble model successfully identifies 48 non-demented and 43 demented cases, it has 2 misclassification cases and 7 unknown demented cases. This model proves to be highly accurate when compared to other independent models like XGBoost because this model not only minimizes error rates but also creates stability among different cases. Though the SVM model shows high accuracy rates, using the ensemble model can be proved more efficient since this model combines two classifiers' strengths.

Model Comparison

Comparative study between the three methods, SVM, XG-Boost, and Ensemble(SX Fusion), has been carried out to determine the efficiency in Alzheimer's detection. The comparison on accuracy of the three models has been depicted in Fig. 5.

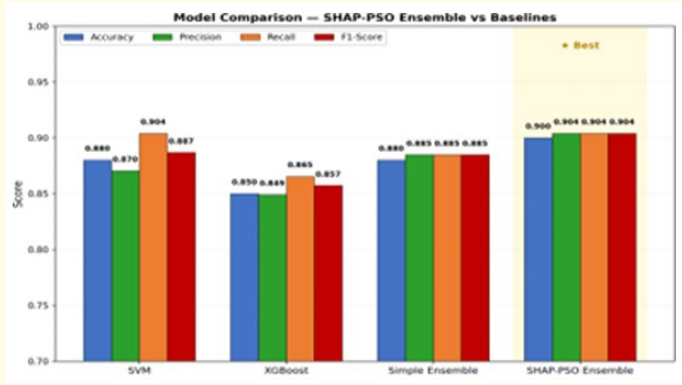


Figure 5: Accuracy Comparison of Models.

Among the three algorithms, SVM classifier has achieved the maximum accuracy and can partition the space through nonlinear decision boundaries, although it may not be able to classify all examples as dementia cases. XGBoost classifier has relatively lower accuracy but can detect the interaction between different features, even though it cannot detect all the cases of dementia, thereby lowering its efficiency. As compared to both the classifiers, SX Fusion model ensures better stability and efficiency of the method, although accuracy is somewhat lower.

In conclusion, it is clear that there cannot be any model that performs well in all aspects. But using the approach of ensemble modeling, the limitations of both models have been well taken care of, increasing its efficiency.

Model	*Accuracy (%)*	*Precision*	*Recall*	*F1-Score*
SVM	0.93	0.96	0.90	0.93
XGBoost	0.89	0.93	0.84	0.88
SX Fusion	0.91	0.95	0.86	0.90

Table 1: Performance Comparison of Models.

MFCC Feature Representation

The Mel Frequency Cepstral Coefficients (MFCC) representation of the speech signal is shown in Fig. 6.

MFCCs represent the transformations taking place in the cepstral coefficients at any specific point in time and thus signify some key spectral characteristics of the speech signal. The cepstral coefficients provide an efficient means for modeling how a person recognizes sounds, emphasizing those frequencies that are important for hearing speech.

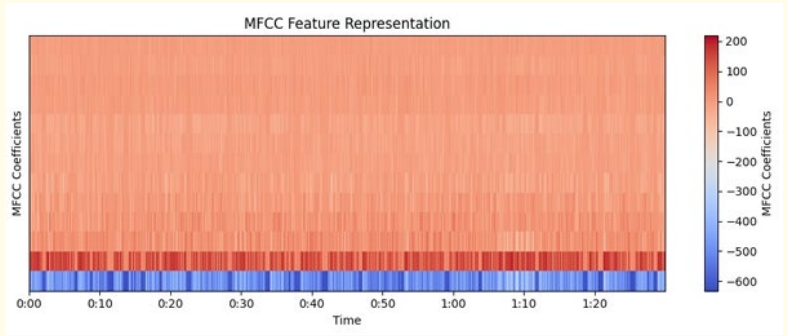


Figure 6: MFCC Feature Representation.

As shown in the above figure, it can be seen that the low order MFCCs represent the spectral envelope of the speech signal, while the high order MFCCs represent the detailed changes in the speech signal.

These changes assist in detecting subtle changes in speech due to impaired cognitive abilities in patients with AD, such as articulation errors and speech patterns.

Because of their informative nature, these features are very suitable for classification using machine learning approaches in the identification of Alzheimer's disease.

Explainability Analysis Using SHAP

To enhance the transparency and understandability of the proposed system, the SHAP technique is used to analyze the contribution of each feature to the XGBoost model. The feature importance plot is depicted in Fig. 7.

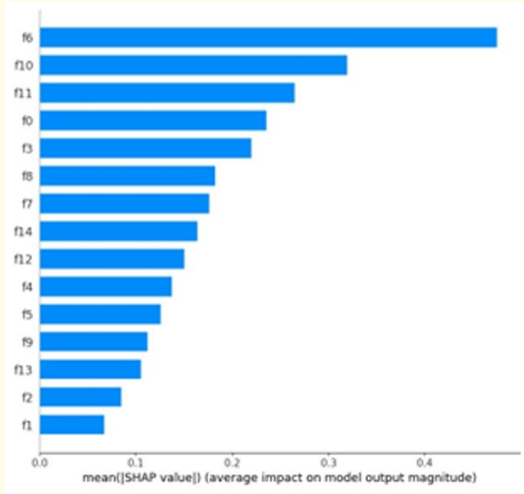


Figure 7: SHAP Feature Importance Plot.

The result from SHAP analysis indicates that there exist some important features that contribute significantly to the prediction model. This suggests that spectral features have significant contributions to discrimination between patients with and without dementia. However, other features like Zero Crossing Rate and Spectral Centroid have no significant roles to play in the decision-making process. They are just supplementary features that serve as inputs to make decisions. From the SHAP value analysis, one can tell how changes in some features affect prediction results.

Discussion

The conclusions drawn from the results of the experiments show that the presented model allows us to find patterns of speech related to the disease. As we can see from the experiment, XGBoost works better than SVM, as XGBoost can deal with the problem of nonlinearity much better. The combination of two models results in enhanced algorithm performance. However, we can say that the results are rather good regarding the number of correct predictions; however, this is caused by the fact that simple parameters were used and not enough data was provided.

Conclusion

The suggested project concerns the development of a speech recognition algorithm to be used for the detection of Alzheimer's disease via such feature extraction algorithms as MFCC, Zero Crossing Rate, and Spectral Centroid as well as SVM and XGBoost. Ensemble learning alongside with SHAP for Explainable AI have been employed to make sure that the suggested approach is reliable. As it follows from the findings, the algorithm can successfully detect features of speech that are associated with the brain functions, and XGBoost and Ensemble learning techniques prove superior to SVM. Despite the fact that accuracy of the model is rather low, it seems promising for further applications in terms of a simple algorithm for detection of Alzheimer's disease. The future research will focus on temporal features of speech and collecting additional speech samples for the database. To sum up, we managed to confirm the viability of the idea.

Future Scope

In addition, the system can be improved by adding extra attributes and models for higher precision and robustness of the system. Some examples of these enhancements may include the integration of new speech attributes from the temporal analysis of speech such as pause duration, speaking speed, and articulation clarity, which would assist in evaluating cognitive impairments. Moreover, the use of pre-trained models employing deep learning algorithms to conduct speech recognition (such as wav2vec or HuBERT) is expected to facilitate the extraction of meaningful semantics and speech features compared to hand-crafted features. In addition, the expansion of the dataset by incorporating more speech samples can lead to higher generalization of the model, minimizing its bias.

References

1. Taherinezhad F., et al. "Speech-Based Cognitive Screening Using Large Language Models". arXiv preprint arXiv:2509.03525 (2025).
2. Li VK., et al. "Reverse-Speech-Finder: Neural Network Backtracking for Alzheimer's Speech Detection". arXiv:2505.17477 (2025).
3. Kiyoshige E., Yamada T and Tanaka H. "AI-Based Voice Biomarkers for Early Detection of Cognitive Impairment". Computer Methods and Programs in Biomedicine (2025).
4. de Filippis R and Al Foysal A. "Early Alzheimer's Detection Using Linguistic Markers from Speech". Open Access Library Journal (2025).
5. Oiza-Zapata I., et al. "Explainable AI for Alzheimer's Detection Using Speech Signals". Electronics 14.11 (2025).
6. Li J., et al. "Benchmarking Foundation Speech Models for Alzheimer's Detection". arXiv:2506.11119 (2025).
7. Park C., et al. "Chain-of-Thought Reasoning for Alzheimer's Detection Using Speech". arXiv:2506.01683 (2025).
8. Zolnoori M., et al. "SpeechCARE: A Multimodal Framework for Alzheimer's Detection". arXiv:2511.08132 (2025).
9. Sharafeldeen A., et al. "Machine Learning Approaches for Speech-Based Alzheimer's Detection: A Review". Computers 14.2

(2025).

10. Rezaii N., et al. "Analyzing Digital Voice for Early Detection of Alzheimer's Disease". Nature Digital Medicine (2025).
11. Samanta H. "Impact of Automatic Speech Recognition Quality on Alzheimer's Detection". arXiv:2603.18239 (2026).
12. Lv J., et al. "Cognitive Acoustic Symbolic Transformation for Alzheimer's Detection". NeuroImage (2026).
13. Wang L., et al. "Multimodal Data Analysis for Early Detection of Alzheimer's Disease". Journal of Prevention of Alzheimer's Disease (2026).
14. Gupta S., et al. "Explainable Machine Learning Models for Speech-Based Dementia Detection". IEEE Access (2025).
15. Sharma P., et al. "Acoustic Feature-Based Alzheimer's Detection Using Ensemble Learning Techniques". IEEE Transactions on Biomedical Engineering (2025).

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